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## NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

### Fifth Semester B.E. (CCE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CC503 / 20CC502 - CRYPTOGRAPHY AND NETWORK SECURITY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II** each and **One full** question from **Unit – III**.*

#### Unit – I

Marks    BT\*    CO\*    PO\*

1. a) Define the following terms:

- i) Encryption algorithm
- ii) Secret key
- iii) Cipher text
- iv) Plain text
- v) Cryptanalysis

5    L\*2    1    1

b) Bring out the differences between monoalphabetic and polyalphabetic ciphers with suitable examples.

5    L2    1    1

c) Consider a Hill cipher with  $m=3$  (block size=3) with key K show below:

$$\begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$$

What is the ciphertext corresponding to the plaintext PAYMOREMONEY? Verify the same by decrypting the one of the cipher text.

10    L3    1    2

2. a) Apply the playfair cipher algorithm for the encryption of the plaintext "CRYPTOGRAPHY". Use the key PLAYFAIR. Also verify the same by decrypting the ciphertext.

10    L3    1    2

b) Elaborate the encryption process in DES algorithm.

10    L2    1    1

3. a) Explain the steps in the encryption and decryption process in RSA algorithm with suitable examples.

10    L2    2    1

b) Alice and Bob use the Diffie–Hellman key exchange technique with a common prime

$q = 157$  and a primitive root  $\alpha = 5$ .

a. If Alice has a private key  $X_A = 15$ , find her public key  $Y_A$ .

b. If Bob has a private key  $X_B = 27$ , find his public key  $Y_B$ .

c. What is the shared secret key between Alice and Bob?

10    L3    2    2

#### Unit – II

4. a) Briefly explain the various approaches for the distribution of public keys.

10    L2    3    1

b) With the help of a neat message format, explain X.509 certificate.

10    L2    3    1

P.T.O.

|       |                                                                       |    |    |   |   |
|-------|-----------------------------------------------------------------------|----|----|---|---|
| 5. a) | Bring out the differences between Kerberos version 4 and version 5.   | 10 | L2 | 3 | 1 |
| b)    | With a neat diagram, explain the key distribution scenario.           | 10 | L2 | 3 | 1 |
| 6. a) | Explain the Secure Socket Layer architecture with a suitable diagram. | 10 | L2 | 4 | 1 |
| b)    | With a neat diagram, explain the SSH protocol stack.                  | 10 | L2 | 4 | 1 |

## Unit – III

|       |                                                                                                     |    |    |   |   |
|-------|-----------------------------------------------------------------------------------------------------|----|----|---|---|
| 7. a) | Explain the transmission and reception of messages of PGP messages through the help of a flowchart. | 10 | L2 | 5 | 1 |
| b)    | Explain with a neat figure, the architecture of IPsec.                                              | 10 | L2 | 5 | 1 |
| 8. a) | Elaborate on S/MIME content types.                                                                  | 5  | L2 | 5 | 1 |
| b)    | Differentiate between PGP and S/MIME.                                                               | 5  | L2 | 5 | 1 |
| c)    | Bring out the differences between the transport mode and tunnel mode.                               | 10 | L2 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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